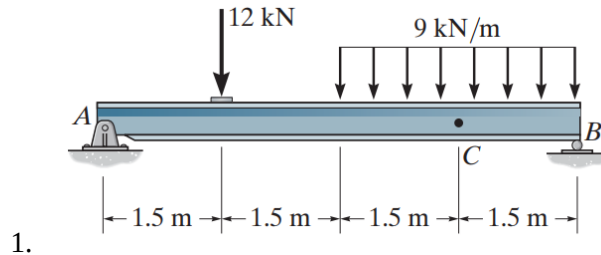


**Tutorial-4**  
**ME-101, (Engineering Mechanics, Semester-II)**  
**Date: 12/02/2024 and Time: 7.55 to 8.50 AM**

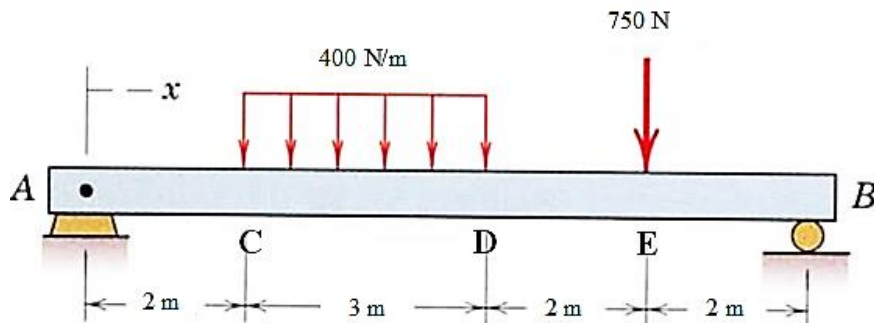
1. Determine the shear force and the bending moment at point C of the beam, shown in **Figure 1**.



**Figure 1**

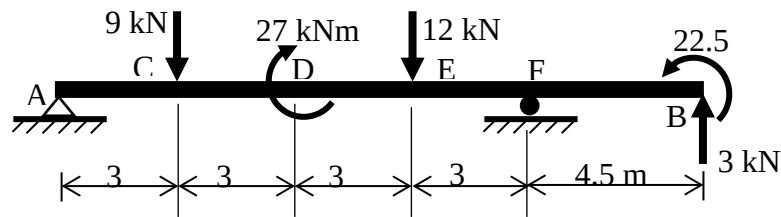
2. **Figure 2** shows a beam loaded with both the distributed and point loads.

- a) Calculate the support reactions at point A and B. [1 mark]
- b) Find the shear force and bending moment at section  $a-a'$ , which is at a distance  $x$  from point A. [1 mark]
- c) Find the shear force and bending moment at section  $b-b'$ , which is at a distance  $x$  from point A. [1 mark]
- d) Find the shear force and bending moment at section  $c-c'$ , which is at a distance  $x$  from point A. [1 mark]
- e) Find the shear force and bending moment at section  $d-d'$ , which is at a distance  $x$  from point A. [1 mark]
- f) Calculate shear force and bending moment at  $x = 6$  m. [1 mark]
- g) Calculate maximum bending moment and its location from point A. [2 marks]
- h) Plot shear force and bending moment diagram for entire beam. [2 marks]



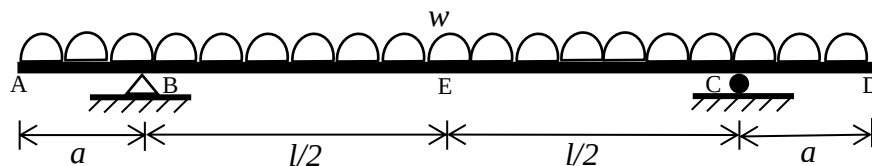
**Figure 2**

3. For the beam and the loading, shown in **Figure 3**, draw the shear force and bending moment diagrams. Determine the maximum values of shear force and bending moment.



**Figure 3**

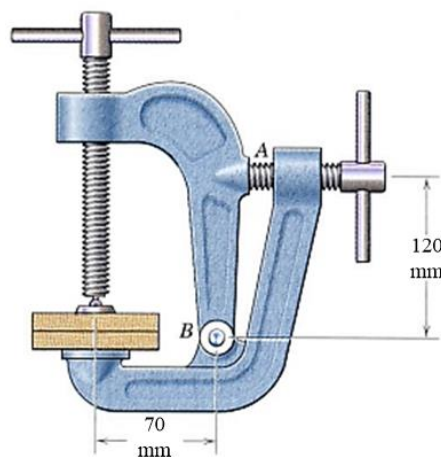
4. A simply supported beam with two equal overhangs is loaded by uniformly distributed load  $w$  on entire length as shown in **Figure 4**. If support position is adjustable, then draw the shear force and bending moment diagrams for following three cases: **a)**  $l > 2a$ ; **b)**  $l = 2a$ ; and **c)**  $l < 2a$ .



**Figure 4**

5. **Figure 5** shows the dual-grip clamp used to provide added clamping force with a positive action. If the vertical screw is tightened to produce a clamping force of 3 kN and then the horizontal screw is tightened until the force in the screw at A is doubled,

- Draw the Free Body Diagram of the clamping system. [3 Marks]
- Find the force acting on the horizontal screw [2 Marks]
- Find the force acting on the vertical screw [2 Marks]
- Find the total reaction  $R$  on the pin at B. [3 Marks]



**Figure 5**